

CS & IT

Theory Of computation

DPP: 1

Regular Language & Grammar

Q1 Consider alphabet $\Sigma = \{a, b\}$, the empty string ϵ and the set of strings S, P, Q and R generated by the corresponding non-terminals of a regular grammar. S, P, Q and R related as follows (S is a start symbol):

$$S \rightarrow aP \mid bQ \mid \epsilon$$

$$P \rightarrow bR \mid aS$$

$$Q \rightarrow aR \mid bS$$

$$R \rightarrow aQ \mid bP$$

$$(A) L = \{w : n_a(w) \text{ and } n_b(w) \text{ both are even}\}.$$

$$(B) L = \{w : n_a(w) \text{ and } n_b(w) \text{ both are odd}\}.$$

$$(C) L = \{w : n_a(w) \text{ or } n_b(w) \text{ are even}\}.$$

$$(D) \text{ None of these.}$$

Q2 Consider the following language L on alphabet $\Sigma = \{a, b\}$
 $L = \{wxw^R \mid w, x \in \{a, b\}^+\}$
 The correct regular grammar of above language is/are possible?

$$(A) S \rightarrow aAa \mid bAb$$

$$A \rightarrow aA \mid bA \mid a \mid b$$

$$B \rightarrow aA \mid bA \mid a \mid b$$

$$(B) S \rightarrow aAa \mid bAb \mid \epsilon$$

$$A \rightarrow ab$$

$$(C) S \rightarrow aA \mid bB$$

$$A \rightarrow aA \mid bA \mid a$$

$$B \rightarrow bB \mid aB \mid b$$

$$(D) S \rightarrow Aa \mid Bb$$

$$A \rightarrow Aa \mid Ab \mid a$$

$$B \rightarrow Bb \mid Ba \mid b$$

Q3 Consider the following grammar G :

G :

$$S \rightarrow A B C$$

$$A \rightarrow aA \mid a$$

$$B \rightarrow bc$$

$$C \rightarrow cC \mid \epsilon$$

The language generated by above grammar is?

$$(A) L = \{a^* bc c^*\}$$

$$(B) L = \{a^+ b c^+\}$$

$$(C) L = \{a^* b c^*\}$$

$$(D) \text{ None of these}$$

Q4 Which of the following language is non-regular?

$$(A) L = \{wxw^R \mid x, w \in \{a, b\}^*\}.$$

$$(B) L = \{wxw \mid w, x \in \{a, b\}^*\}.$$

$$(C) L = \{wxwx \mid w, x \in \{a, b\}^*\}.$$

$$(D) \text{ None of these}$$

Q5 Consider the following grammars G_1 and G_2 :

G_1 :

$$S \rightarrow aAb$$

$$A \rightarrow aB \mid \epsilon$$

$$B \rightarrow Ab$$

G_2 :

$$S \rightarrow aABb$$

$$A \rightarrow aA \mid \epsilon$$

$$B \rightarrow bB \mid \epsilon$$

Which of the following grammar is/are regular?

$$(A) G_1 \text{ only}$$

$$(B) G_2 \text{ only}$$

$$(C) \text{ Both } G_1 \text{ and } G_2$$

$$(D) \text{ None of these}$$

Q6 Consider the following three languages:

$$(1) L = \{a^{n^2} \mid n \geq 1\}$$

$$(2) L = \{a^{m^2} \mid m = n^2, n \geq 1\}$$

$$(3) L = \{a^{mn} \mid n \geq 1, m > n\}$$



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Total number of regular languages is/are_____.

(D) None of these

Q7 Which of the following language is non-regular?

(A) $L = \{a^{2m} b^n b^n \mid m, n \geq 1\}$

(B) $L = \{a^m b^n X \mid m, n \geq 1, X \in \{a, b\}^*\}$

(C) $L = \left\{ \left\{ a^{n^2} \right\}^* \mid n \geq 0 \right\}$

(D) None of these

Q8 Consider following statements:

S_1 : Kleene Closure (*) of infinite set is always finite.

S_2 : Kleene Closure (*) of finite set is always infinite.

Which of the following is correct?

(A) S_1 only

(B) S_2 only

(C) Both S_1 and S_2

(D) None of these

Q9 Consider the following statements:

[I] If L is regular, then $\bar{L}$ is regular.

[II] If $\bar{L}$ is regular, then L is regular.

[III] Union of L and its complement is Σ^*

Number of correct statement is/are_____.

Q10 Consider a regular language L , which of the following statements are true regarding L .

(A) $\text{Prefix}(L) = \{w \mid ww_1 \in L, w_1 \in \Sigma^*\}$ is regular.

(B) $\text{Suffix}(L) = \{w \mid w_1 w \in L, w_1 \in \Sigma^*\}$ is regular.

(C) $\text{Quotient}(L)$ is regular.

(D) L is closed under infinite intersection.

Q11 Consider a regular language L over the alphabet

$\Sigma = \{a, b\}$. L is defined as $L = (a + b^*)(bab^*)$.

If homomorphism h is defined over $T = \{c, d, e\}$ and

$h(a) = cd$

$h(b) = cddec$

Then the regular language $h(L)$ is given as

(A) $(cd + cddec)(cddec cd cddec)$

(B) $(cddec)(cd + cddec^*)$

(C) $(cd + (cddec)^*)((cddec)(cd)(cddec)^*)$



Answer Key

Q1 (A)

Q2 (A)

Q3 (B)

Q4 (D)

Q5 (B)

Q6 1

Q7 (D)

Q8 (D)

Q9 3

Q10 (A, B, C)

Q11 (C)



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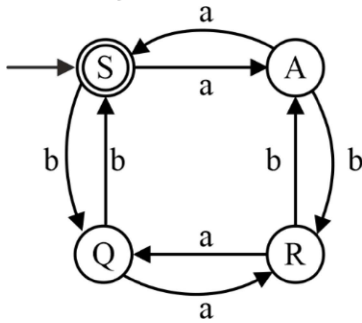
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Hints & Solutions

Q1 Text Solution:

DFA for grammar:



$$L = (aa + ab + ba + bb)^*$$

Hence, option (a) is correct.

Q2 Text Solution:

$$L = \{wxw^R \mid w, x \in \{a, b\}^+\}$$

$$a(a+b)^+a \mid b(a+b)^+b$$

↓ ↓

$$ab(a+b)^+ba \mid ba(a+b)^+ba$$

$$aa(a+b)^+aa \mid bb(a+b)^+bb$$

$L = \text{Regular}$

Regular expression = $a(a+b)^+a + b(a+b)^+b$

$$S \rightarrow aAa \mid bAb$$

$$A \rightarrow aA \mid bA \mid a \mid b$$

$$B \rightarrow aA \mid bA \mid a \mid b$$

Regular expression for above grammar is $a(a+b)^+a + b(a+b)^+b$

Hence, only (a) is correct.

Q3 Text Solution:

$$S \rightarrow ABC = aa^*bcc^*$$

$$A \rightarrow aA \mid a = aa^*$$

$$B \rightarrow bc = bc$$

$$C \rightarrow cC \mid \epsilon = c^*$$

$$\text{Regular expression} = aa^*bcc^*$$

$$= a^+bc^+$$

Hence, option (b) is correct.

Q4 Text Solution:

(a) $L = \{wxw^R \mid x, w \in \{a, b\}^*\}$

$$\text{Minimal string} = \epsilon \cdot (a+b)^* \epsilon$$

$$= (a+b)^*$$

Regular

(b) $L = \{wxw \mid w, x \in \{a, b\}^*\}$

$$L = \epsilon \cdot (a+b)^* \cdot \epsilon$$

$$= (a+b)^*$$

Regular

(c) $L = \{wxwx \mid w, x \in \{a, b\}^*\}$

regular

Hence option (d) is correct.

Q5 Text Solution:

Only G_2 is regular.

Q6 Text Solution:

(1) $L = \{a^{n^n} \mid n \geq 1\}$

$$L = \{a, a^4, a^{27}, \dots\}$$
 Non-regular

(2) $L = \{a^{m^n} \mid m = n^2, n \geq 1\}$

$$L = \{a^{1^1}, a^{4^2}, a^{9^3}, \dots\}$$

$$= \{a, a^{16}, a^{43}, \dots\}$$

Non-regular

(3) $L = \{a^{m^n} \mid n \geq 1, m > n\}$

$$L = \{a^{2^1}, a^{3^1}, a^{4^1}, \dots\}$$

$$= \{a^2, a^3, a^4, \dots\}$$

$$= aa(a)^*$$

Regular

Q7 Text Solution:

(a) $L = \{a^{2^m}b^n \mid m, n \geq 1\}$

$$= (aa)^+b^{2n}$$

$$= (aa)^+(bb)^+ \text{ Regular}$$

(b) $L = \{a^m b^n X \mid X \in \{a, b\}^*, m, n \geq 1\}$

$$= (a)^+(b)^+(a+b)^*$$

= Regular

(c) $L = \left\{ \left\{ a^{n^2} \right\}^* \mid n \geq 0 \right\}$

$$L = \{\epsilon, a, aa, aaa, \dots\}$$

$$= a^*$$



= Regular

Hence, option (d) is correct.

Q8 Text Solution:

S₁: False

Set = $\{\epsilon\} = \{\epsilon\}^* = \epsilon$ only (Finite)

S₂: Set = $\{a\} = \{a\}^* = \epsilon, a, aa, aaa, \dots = (a^*)$ (Infinite)

So, both statements are false.

Hence, option (d) is correct.

Q9 Text Solution:

- L is regular if and only if Complement of L is regular.

- $L \cup \bar{L} = \Sigma^*$

Hence, all are correct statements.

Q10 Text Solution:

Regular language is closed under Prefix, Suffix and quotient of the language. But regular language are not closed under infinite intersection.

So, a, b, c are correct.

Q11 Text Solution:

Homomorphism is a function from strings to string which is based on concatenation.

for any a and b

L is defined as

$x = (a + b)^* (bab)^*$

then,

$h(L) = (h(a) + h(b))^* (h(b)h(a)h(b))^*$

$= (cd + (cddec)^*) ((cddec)(cd)(cddec)^*)$.



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